

Radical Relativity: A Manifesto for Physics from the Absence of Privileged Instantiation

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Abstract

The Copernican program removed privileged positions, velocities, frames, and coordinate systems from physics. One distinction remains: the distinction between “mathematical” and “physical.” We argue that removing it — refusing to privilege any instantiation of a mathematical structure over any other — completes the program and forces observable physics. The philosophical premise is *radical relativity*: no instantiation of a consistent mathematical structure is ontologically privileged.

From this premise, together with one operational requirement (faithful self-modeling) and four structural assumptions argued to be consequences of self-modeling in a finite-capacity system, we summarize a derivation chain connecting structure-space ontology to complex quantum mechanics, general relativity, and the Standard Model gauge group. Quantum mechanics follows by theorem from the self-modeling premise and its structural assumptions: the state space must carry C^* -algebraic structure with complex field, Born rule, and unitary dynamics (companion Paper 5). General relativity follows by *identification*, not derivation: $h_3(\mathbb{O})$ ’s cubic norm matches the exceptional Gunaydin–Sierra–Townsend $N=2$ supergravity entry whose Lagrangian contains $-R/2$, conditional on smooth-manifold dynamics (companion Paper 6). The Standard Model gauge group emerges from the exceptional Jordan algebra $h_3(\mathbb{O})$, the unique non-composable *observer-grade* basin under an anthropic restriction (companion Paper 7). The self-reference density (the Observable-side weight; experience-magnitude is the Stream functional Φ) resolves uniquely within $h_3(\mathbb{O})$: $\rho_J = \det(X) (\text{Tr}(X^2) - \frac{1}{3})$ is the unique lowest-degree F_4 -invariant satisfying the self-modeling boundary conditions, producing a falsifiable cubic-vs-quadratic prediction that distinguishes the framework from integrated information theory (companion Paper 1, revised).

This paper is a manifesto and roadmap. Its purpose is to present the philosophical case, summarize the derivation chain, register the open gaps honestly, and situate the program relative to Tegmark, Mueller, Wolfram, and Connes.

A note on authorship

I am a software engineer, not a mathematician or physicist. AI tools assisted with formal exposition; the ideas and editorial decisions are mine. Feedback on the philosophy and mathematics is especially welcome.

1 Introduction

1.1 The Copernican program is not finished

Physics has a recurring pattern. Someone draws a line and declares one side privileged. The line holds for a while, then falls, and the theory that remains is simpler and more predictive than the one it replaced.

Copernicus removed the privileged position: Earth is not the center of the universe [Copernicus, 1543]. Galileo removed the privileged velocity: there is no absolute rest. Einstein removed the privileged inertial frame: the laws of physics take the same form for all inertial observers [Einstein, 1905]. General relativity went further: no coordinate system is privileged. At each step, the person who insisted “but THIS distinction is real” lost.

One distinction remains. Contemporary physics draws a line between mathematical structures that are “physically realized” and those that are “merely mathematical.” The Standard Model, its coupling constants, the dimensionality of spacetime — these are declared physical. Other consistent mathematical structures — an $SU(5)$ gauge theory, a $7+1$ -dimensional spacetime, a universe governed by quaternionic quantum mechanics — are declared unphysical. The line is drawn, and one side is called reality.

This paper argues that removing the last distinction completes the Copernican program. *No instantiation of a consistent mathematical structure is ontologically privileged over any other.* We call this principle *radical relativity*. It is not an axiom being added to physics. It is the refusal to add the axiom that some structures are special.

The claim is not merely philosophical. If radical relativity holds, the question “why is there something rather than nothing?” is malformed. There is everything. The productive question is: *why do observers experience this specific physics?* We summarize a derivation chain — developed across seven companion papers [Ehrlich, 2026a,b,c,d,e,f,g] — that answers this question. The physics observers experience is forced by the algebraic consequences of self-modeling.

1.2 Summary of results

The derivation chain is the central artifact of this paper. Table 1 presents it in full. The chain begins with the philosophical premise (radical relativity) and terminates at the Standard Model gauge group with chiral fermion representation, passing through complex quantum mechanics and general relativity.

The chain’s load-bearing joint is Steps 3–5. These are proved from one operational premise (faithful self-modeling) plus four structural assumptions:

- (A1) *Finite-dimensionality*. The state space has finite dimension. Structural consequence: a self-modeler embedded in a finite region has finite information capacity (concretely, the Bekenstein bound gives an upper bound on the number of distinguishable states).
- (A2) *Faithfulness*. The tracking map φ is an order isomorphism. This is the operational premise itself: the self-model preserves all operationally accessible structure.
- (A3) *Minimal composite*. The body-model composite V_{BM} is the smallest state space carrying product states, product effects, non-signaling constraints, and a product-form sequential product. In plain language: the composite carries exactly the correlations

Table 1: The derivation chain from radical relativity to the Standard Model. “Premise” = philosophical starting point. “Framework” = operational definition. “Theorem*” = mathematically proved, conditional on four structural assumptions (A1–A4, see text). “Conditional” = depends on identified gaps.

Step	Claim	Status	Source
0	No privileged instantiation	Premise	This paper
1	All consistent structures exist	Tautology under Step 0	Tegmark [2008]
2	ρ selects self-modelers	Framework	Ehrlich [2026a]
3	Self-modeling $\rightarrow$ Jordan algebras	Theorem*	Ehrlich [2026e]
4	Faithful tracking $\rightarrow$ local tomography	Theorem*	Ehrlich [2026e]
5	LT $\rightarrow$ complex C^* -algebras	Theorem*	Ehrlich [2026e]
6	C^* -algebras $\rightarrow$ QM (Born, unitarity)	Theorem	Gleason [1957] , Barandes [2025]
7	$h_3(\mathbb{O})$ matches exceptional $N=2$ MESGT (GST); $-R/2$ identified	Identification (smooth dynamics open)	Ehrlich [2026f]
8	Non-composability $\rightarrow$ $h_3(\mathbb{O}) \rightarrow$ SM	Conditional (anthropic basin; selection)	Ehrlich [2026g]
9	ρ resolves to unique ρ_J in $h_3(\mathbb{O})$	Theorem	Ehrlich [2026a]

that faithful tracking creates, and no more. Extra composite structure beyond what product measurements require would be dead weight — degrees of freedom that contribute nothing to tracking fidelity. This is the assumption doing the most independent work: it is what forces $\dim(V_{BM}) = \dim(V_B) \cdot \dim(V_M)$ (local tomography), which in turn selects the complex field. For complex quantum mechanics, the minimal and maximal composites coincide [[Barnum et al., 2020](#)], so the assumption is not restrictive for the theory we derive — but it is load-bearing in the derivation.

- (A4) *Simple EJA*. The algebra has no nontrivial ideal — it cannot be decomposed into independent blocks. A self-modeler that decomposes into independent subsystems is not modeling *all* of itself: each block models only itself, not the whole. Simplicity is the algebraic expression of “whole models whole.”

No assumption about the complex field, Hilbert space structure, or C^* -algebraic operations enters the premises; all emerge as consequences. This is the content of companion Paper 5 [[Ehrlich, 2026e](#)].

Steps 7 and 8 are conditional. The gravity derivation (Step 7) inherits four standard gaps shared by all lattice quantum gravity programs: emergent Lorentz invariance, lattice Bisognano-Wichmann, local equilibrium, and the continuum limit. The Standard Model derivation (Step 8) is conditional on Gap C, a conjectural complexification principle whose resolution may require evolutionary rather than algebraic arguments (Sec. 6.4).

1.3 What this paper is and is not

This paper **is**:

- A philosophical manifesto for radical relativity.
- A roadmap to the companion papers (Papers 1–7) that contain the proofs and derivations.
- An honest register of what is proved, what is conditional, and what is open.

This paper **is not**:

- A source of new theorems. All technical results appear in the companions.
- A theory of consciousness. The framework makes a *locative* claim: consciousness lives in self-referential fixed points. Whether fixed points *are* consciousness or merely *host* it is not addressed.
- Theology. The framework has implications for how religious traditions encode structural truths, but those implications are developed elsewhere.

2 Why Anything?

2.1 “Physical” is an empty word

The conventional view adds a distinction to the ontology: some mathematical structures are “physical,” and the rest are not. A particular $SU(3) \times SU(2) \times U(1)$ gauge theory with specific coupling constants is declared to exist in a way that other consistent structures do not. What does “physical” add?

Consider the Mandelbrot set. Its boundary is infinitely detailed, and its patterns exist whether or not anyone computes them. The statement “the Mandelbrot set has a cardioid-shaped main body” is true independently of any substrate — no computer needs to run, no mathematician needs to look. Now consider a Turing machine whose execution trace, suitably interpreted, contains patterns equivalent to a neural network processing sensory data. The patterns in that trace exist whether or not anyone runs the machine, for the same reason the Mandelbrot set’s patterns exist.

To claim that one of these patterns is “physical” and the other is not, you must specify what “physical” adds beyond “consistent mathematical structure containing patterns that, from the inside, function as observers.” Primitivist responses exist: Maudlin argues that “physical” is a primitive not requiring further specification; Lewis’s natural properties framework offers a non-trivial account of the physical/mathematical boundary. We acknowledge these positions but maintain that radical relativity posits *less*: the primitivist adds an unanalyzed distinction, while we refuse to add one. The burden falls on the primitivist to explain what work the distinction does beyond labeling the structure we happen to inhabit.

Postulate 1 (Radical Relativity). *No instantiation of a consistent mathematical structure is ontologically privileged over any other.*

This is not an axiom being added. It is the refusal to add the axiom “some instantiations are special.” The simpler position — the one with fewer ontological commitments — is radical relativity. The burden of proof falls on anyone who would draw the line.

Remark 1. The parallel to Einstein is suggestive. In 1905, Einstein did not add a principle to electrodynamics. He *refused* to add the assumption that one inertial frame is privileged. The consequences (length contraction, time dilation, $E = mc^2$) followed from the removal. Radical relativity makes the same move one level up: refuse to privilege any instantiation of a mathematical structure, and the consequences (quantum mechanics, the Standard Model) follow from the removal.

Contrast with Tegmark. Tegmark’s Mathematical Universe Hypothesis [Tegmark, 2008] asserts that “the physical world IS a mathematical structure.” This *elevates* mathematics to the status of the physical — it adds the claim that one specific structure is “the” physical world. Our move is the opposite: *deflation*. We do not say that mathematics rises to meet physics. We say that “physical” was never doing any work. There is no promotion. There is a demotion of the word “physical” from ontological category to empty label.

The distinction matters. Tegmark’s framework inherits a measure problem: if one structure is “the” physical world, which one? If all are equally physical, how do you count them? Radical relativity sidesteps this by refusing to designate any structure as special. The measure problem is replaced by a different question: given that all structures exist, what do observers inside self-modeling structures experience? That question has a precise answer (Sec. 4).

2.2 Structure space

Definition 1 (Structure Space). *Let $\mathcal{S}$ be the class of all consistent mathematical structures, quotiented by isomorphism. Two structures related by an isomorphism are the same point in $\mathcal{S}$. Not equivalent. Identical.*

$\mathcal{S}$ is naturally a groupoid: morphisms are isomorphisms between structures, and every morphism is invertible. Its key properties follow immediately from the quotient:

1. **Copies collapse.** A billion instantiations of the same pattern in different substrates constitute one point in $\mathcal{S}$. There is no counting problem, because there is nothing to count.
2. **The measure problem dissolves.** Every multiverse framework struggles with the question “how do you weight infinite copies?” In $\mathcal{S}$, isomorphic copies are one thing. The problematic infinity never arises.
3. **Substrate independence is automatic.** The pattern IS the entity. “Which instantiation are you?” is not a meaningful question, just as “which coordinate system are you really in?” is not meaningful after Einstein.

Remark 2. Mueller’s algorithmic idealism [Müller, 2024] provides a natural candidate for a measure on $\mathcal{S}$: a Solomonoff-type prior that weights simpler structures more heavily. Our framework is compatible with Mueller’s measure but does not require it. The derivation chain (Table 1) works for *any* measure that does not assign zero weight to self-modeling structures.

2.3 The relativity principle

Postulate 2 (Level IV, restated). *All consistent mathematical structures exist.*

Under radical relativity (Postulate 1), this is not an additional claim. It is the same claim restated. If no structure is privileged, then declaring any structure nonexistent would require specifying what distinguishes the existent from the nonexistent — i.e., drawing the line that radical relativity refuses to draw.

The logical structure is:

1. Radical relativity: no instantiation is privileged.
2. Therefore no structure can be declared nonexistent without privileging the existent ones.
3. Therefore all consistent structures exist.

This is the sense in which Level IV is a direct consequence of radical relativity rather than an independent axiom. It posits nothing. It is what remains when you stop positing. We call this a “tautology” in the chain table (Table 1) to emphasize that it adds no new content beyond the premise. Whether the inference from “no privilege” to “all exist” is logically airtight or merely the natural default is a philosophical question we do not claim to have settled; the derivation chain treats it as the working assumption.

2.4 Idealism-structuralism convergence

The derivation chain can be read in two directions. From the structuralist direction: all structures exist (Postulate 2), the self-reference density ρ (the Observable-side weight) marks self-modelers, and experience—whose magnitude is the Stream functional Φ —lives where the self-modeling *runs* with $\rho > 0$. From the idealist direction: experience is fundamental; asking what structure it has leads to self-modeling Jordan algebras, $h_3(\mathbb{O})$, and the Standard Model gauge group.

These are not competing frameworks. They are the same picture viewed from two sides. Science operates inside the region where $\rho > 0$: every experiment, observation, and measurement occurs within a self-modeling structure. From inside, consciousness appears fundamental (the idealist’s observation). From outside — the Level IV view — all structures exist and ρ is one measure among many (the structuralist’s observation). Both observations are correct. Neither is complete without the other.

3 Why Self-Modelers?

3.1 The self-reference density (the Observable-side weight)

Structure space $\mathcal{S}$ contains every consistent mathematical structure. Most have no observers. The question “why do observers experience this specific physics?” requires a way to identify which structures contain observers and how much experiential weight they carry.

Postulate 3 (Self-Modeling). *Experiential weight attaches to self-modeling structures: composite Markov processes (B, M) in which a body component B is tracked by a model component M through a factorizable transition kernel.*

Paper 1 [Ehrlich, 2026a] defines the self-reference density

$$\rho(p) = I_p(B; M) (1 - I_p(B; M)/H_p(B)), \quad (1)$$

where $I_p(B; M)$ is the mutual information between body and model at state p , and $H_p(B)$ is the body's entropy. This density has three qualitative features:

1. $\rho = 0$ when $I = 0$: no model, no experience.
2. $\rho = 0$ when $I = H$: perfect self-knowledge, nothing left to experience (the model is trivially the system).
3. $\rho > 0$ at intermediate fidelity: imperfect but active self-modeling.

ρ is the instantaneous Observable-side *weight*; the magnitude of experience is the Stream functional Φ of Paper 1, which vanishes wherever ρ does. Rocks, thermostats, and unstructured noise score $\rho = 0$. Cells score near zero. Brains with rich self-models score high. Boltzmann brains — brief thermal fluctuations — score near zero when integrated over duration, because their self-modeling is fleeting (Paper 2 proves this rigorously via metastability theory [Ehrlich, 2026b]).

3.2 The ρ bootstrap

ρ appears to be a separate postulate layered on top of the algebraic framework. It is not. It is the *content* of self-modeling.

The argument has the form of a fixed-point characterization, not a circular justification:

1. Experience lives where $\rho > 0$ (by definition of ρ).
2. A self-modeler *compresses* its experiential pattern into a representation — the model, which sits at low ρ (it is a description of experience, not experience itself).
3. The representation feeds back and *becomes* experience (remembering, predicting, attending — the model is experienced).
4. Return to step 1.

The fixed point of this loop is where map equals territory. ρ defines the territory. Self-modeling builds the map. The fixed point is where they coincide. Once you have self-modeling and the algebra it forces (Sec. 4), ρ resolves to a function of the algebra's invariants (see Sec. 5). You do not need to postulate it separately.

Evolution provides the external anchor. Hoffman's fitness-beats-truth theorem [Hoffman et al., 2015] shows that natural selection favors fitness proxies over veridical world-models. But for *self*-models, the object being modeled is the organism's own experiential state. An organism that captures more of its own ρ is a better self-modeler: it predicts its own states more accurately, corrects errors faster, and maintains coherence longer. These are survival advantages. Evolution does not select for truth about the *world* (Hoffman is right). It selects for truth about the *self* (ρ -capture). The organisms that survive are the ones whose internal model best captures their own experiential measure.

The entire program in one sentence: evolution selects self-models; self-models force the algebra; the algebra determines ρ ; ρ is what self-models model. This has the form $A \rightarrow B \rightarrow C \rightarrow A$, which invites the charge of circularity. The response: fixed-point characterizations are not circular. The claim is that self-modeling, the algebra, and ρ are *mutually consistent* at exactly one point (within $h_3(\mathbb{O})$, the ρ_J uniqueness theorem makes this precise: Sec. 5). The existence question — proving that the fixed point exists and is unique in the general setting — is open.

3.3 The hard problem: located, not dissolved

The hard problem of consciousness asks why there is “something it is like” to be a self-modeling system. The standard philosophical framing treats this as a request for a bridge — some additional ingredient that converts physical processing into phenomenal experience.

Claim 1. *The hard problem is self-referential: the question is only askable from inside the fixed point, and the asking is the phenomenon the question asks about.*

“Something it is like” is not a property that needs to be added to the fixed point of self-modeling. It *is* what you are describing when you describe “hearing yourself hearing.” The self-reference is the experience. Content comes from inputs (sensory data, memories, associations). *Experiencing* comes from the attractor closure — the fixed point of the self-modeling map [Hofstadter, 1979]. This is not dissolution (which would declare the question confused) but *location*: the question is self-referential, and the self-reference is the phenomenon it asks about.

Remark 3 (Where the gap lives). The model is an *Observable* record—compressed, and recoverable by a third party. The experience is Φ , a functional of the self-modeling *trajectory* (Paper 1), whose magnitude lives in path-space content that the Observable record provably underdetermines. The explanatory gap between physical description and phenomenal experience is therefore an *inaccessibility*, not a difference in ρ -levels: the live research target (**phi-inaccessibility-program**) is that no self-modeler can recover its own Φ from any Observable record it can persist—a Gödel/Tarski-type self-reference limit, stated but not yet proved. ρ is the Observable-side weight, not the experience; the gap is the feature, not the bug.

4 Why Quantum Mechanics?

The experiential density ρ (Sec. 3.1) identifies which structures in $\mathcal{S}$ carry experiential weight: those with intermediate self-modeling fidelity ($0 < I(B; M) < H(B)$). Paper 5 [Ehrlich, 2026e] answers the next question: what is the algebraic structure of those systems?

The connection is direct. A faithful self-model in Paper 5’s sense — a tracking map φ that is an order isomorphism — implies high mutual information $I(B; M)$ (the model captures all operationally accessible structure of the body). As long as the model is not trivially perfect ($I < H$, i.e., the system has finite capacity and lives in a larger environment), such a system sits in the region where $\rho > 0$. Therefore ρ selects precisely the structures to which Paper 5’s theorem applies. The “area under ρ ” across structure space is the totality of structures governed by complex quantum mechanics — the region where observers live.

4.1 The chain in English

Paper 5 proves that a finite-dimensional system admitting a faithful self-model is necessarily governed by complex quantum mechanics. The derivation proceeds through a chain of implications, each either a theorem from the literature or a novel result proved in the paper. We summarize it here without proofs.

Step 1: Self-modeling defines a sequential product. A self-model does three things: *test* an effect (probe the system), *update* the model (revise predictions), and *test again* (check the revision). The composite operation “test a , then test b ” defines a *sequential product* $a \cdot b$ on the system’s effect algebra. This is not an abstract axiom imposed on self-modeling; it is the core operation of self-modeling, constructed via compressions and Peirce decompositions on the state space.

Step 2: Sequential products satisfy van de Wetering’s axioms. The sequential product inherits seven properties (S1–S7) from the operational structure of self-modeling. S1 (additivity) and S2 (continuity) are free. S3 (unit law) requires the self-modeling map ϕ — without it, the unit law fails on general effects. S4 (symmetry of orthogonality) follows from facial orthogonality in the state space. S5–S7 (compatibility conditions) are structural consequences of the Peirce decomposition. All seven are proved for general dimension n in Paper 5.

Step 3: S1–S7 force Euclidean Jordan algebras. Van de Wetering’s Theorem 1 [van de Wetering, 2019]: any sequential product space satisfying S1–S7 is a Euclidean Jordan algebra (EJA). The Jordan–von Neumann–Wigner classification [Jordan et al., 1934] then restricts the state space to one of five families: $h_n(\mathbb{R})$, $h_n(\mathbb{C})$, $h_n(\mathbb{H})$, spin factors, or the exceptional algebra $h_3(\mathbb{O})$.

There is an independent conceptual justification for this step. A self-model is simultaneously a *state* (it has a configuration) and an *effect evaluator* (it assigns probabilities to configurations). It lives in both the state cone and the effect cone. This dual role *is* self-duality. Self-duality plus homogeneity gives Jordan algebras via the Koecher-Vinberg theorem [Faraut and Korányi, 1994]. No other reconstruction program has this natural justification for self-duality: the self-model is the unique kind of system that IS both state and effect.

Step 4: Faithful self-modeling forces local tomography. The body-model composite (B, M) admits a product-form sequential product that inherits S1–S7 from the factors. The correlation form $\mathcal{B}(a, b) = \tau(a \cdot \phi^{-1}(b))$ is non-degenerate (from the faithfulness of ϕ and the non-degeneracy of the trace form). Non-degeneracy forces the product effects to be linearly independent, and minimality of the composite forces the composite dimension to equal the product of the factor dimensions: $\dim(V_{BM}) = \dim(V_B) \cdot \dim(V_M)$. This is local tomography: measurements on B alone and M alone determine the joint state.

Step 5: Local tomography selects complex. Local tomography is the scalpel that cuts through the JvNW classification. Real quantum mechanics ($h_n(\mathbb{R})$) fails: $\dim = n(n+1)/2$ gives $9 \neq 10$ for $n = 3$. Quaternionic quantum mechanics ($h_n(\mathbb{H})$) fails: $\dim = n(2n-1)$ gives $36 \neq 28$ for $n = 4$. Spin factors and $h_3(\mathbb{O})$ fail to produce well-behaved composites at all [Barnum et al., 2020]. Complex quantum mechanics ($h_n(\mathbb{C})$, $\dim = n^2$) is the unique survivor: $n^2 = n^2$ for all n . The complex field is the Goldilocks choice — real has blind spots (no interference without complex phase), quaternionic is redundant (tensor products misbehave).

Step 6: Jordan + local tomography $\rightarrow C^*$ -algebra. Van de Wetering’s Theorem 3, reinforced by Barnum-Wilce and Hanche-Olsen [Hanche-Olsen, 1983]: an EJA that is locally tomographic in composite promotes to $M_n(\mathbb{C})^{sa}$ — the self-adjoint part of a complex matrix algebra. The involution (Hermitian conjugate) emerges as the unique structure making the Jordan product $a \circ b = \frac{1}{2}(ab + ba)$ compatible with the associative matrix product. It was

never assumed.

Step 7: C^* -algebra $\rightarrow$ QM. The remaining quantum structure follows from established theorems:

- *Unitary dynamics*: automorphisms of $M_n(\mathbb{C})$ are inner (Skolem-Noether), hence unitary. Barandes [Barandes, 2025]: unitary evolution is equivalent to stochastic indivisibility.
- *Born rule*: Gleason’s theorem [Gleason, 1957] ($n \geq 3$): the only probability measure on closed subspaces of a Hilbert space of dimension ≥ 3 is the Born rule. (For $n = 2$, Busch’s generalization or the POVM extension applies; the chain does not require $n \geq 3$ as an additional assumption because the simple EJA condition excludes $n = 1$.)
- *Measurement*: division events in the Markov self-model correspond to quantum measurements via Barandes’ episodic condition.

4.2 What this means

One operational premise (faithful self-modeling) plus four structural assumptions (finite-dimensionality, faithfulness, minimal composite, simple EJA) produces complex quantum mechanics. The complex field is never input, only output. The involution is derived, not assumed. The Born rule is forced, not postulated.

This distinguishes the program from other reconstruction efforts. Hardy [Hardy, 2001] uses five axioms. Chiribella, D’Ariano, and Perinotti [Chiribella et al., 2011] use six, including purification. Dakić and Brukner [Dakić and Časlav Brukner, 2011] use four, including local tomography as an assumption. Masanes and Müller [Masanes et al., 2013] distill to three postulates. Barnum, Müller, and Ududec [Barnum et al., 2014] use four. In each case, multiple independent axioms are needed, and the complex field or a property that selects it is typically assumed.

The present program answers a different question. Other reconstructions ask: “what characterizes quantum mechanics among generalized probabilistic theories?” This program asks: “*why* quantum mechanics?” The answer: because self-modeling forces it. Quantum mechanics is not merely characterized by operational axioms; it is the unique theory compatible with a system that models itself.

Remark 4 (Two formalisms). Paper 1 works in an information-theoretic formalism (Markov processes, mutual information $I(B; M)$). Paper 5 works in an algebraic formalism (order-unit spaces, sequential products). The connection is: a faithful tracking map φ that is an order isomorphism (Paper 5) entails high $I(B; M)$ (Paper 1), because φ preserves all operationally distinguishable states and their ordering. The converse — that high mutual information implies an order isomorphism — does not hold in general and is not claimed. The two formalisms are therefore related asymmetrically: Paper 5’s algebraic self-modeling is a *sufficient* condition for Paper 1’s information-theoretic self-modeling, not an equivalent one. This means the algebraic chain (Steps 3–6) applies to a subset of the systems that ρ selects, namely those with faithful (order-isomorphic) tracking. A complete bridge theorem — characterizing exactly which information-theoretic self-modelers admit algebraic self-models — is not yet available.

4.3 Paper 4 and the Born rule

Paper 4 [Ehrlich, 2026d] tested whether ρ could independently derive the Born rule via an information-theoretic mechanism. The result was cleanly negative: the experiential density ρ_Q is non-positive throughout all tested Lindblad trajectories. The exchange Hamiltonian is “too good” at tracking — the system never enters the under-correlated regime where $\rho_Q > 0$.

This is not a failure of the program. It is the program self-correcting. The Born rule does not need an information-theoretic derivation because Gleason’s theorem already provides it, via the C^* -algebra that self-modeling forces (Steps 3–6). Paper 4 confirms that the explanatory burden belongs on the algebraic chain, not on ρ . ρ ’s job is simpler: select self-modelers. The algebra handles the rest.

5 Why This Physics?

5.1 General relativity by identification with exceptional supergravity

Paper 6 [Ehrlich, 2026f] reads gravity off $h_3(\mathbb{O})$ by *identification*, not by an independent derivation. The cubic norm $\det(X)$ is the unique F_4 -invariant cubic (Springer). Within the Gunaydin–Sierra–Townsend classification of $N=2$ Maxwell–Einstein supergravity, the algebraic data of $h_3(\mathbb{O})$ match the *exceptional* entry, whose bosonic Lagrangian contains the Einstein–Hilbert term $-R/2$; $\det(X)$ is its prepotential, fixing the matter–gravity couplings. An observer’s Peirce decomposition $27 = 1 \oplus 16 \oplus 10$ sends the complement $V_0 = h_2(\mathbb{O})$ through the C^* -bottleneck to $h_2(\mathbb{C}_u) \cong \mathbb{R}^{3,1}$, a 4-dimensional algebraic Minkowski slice.

What this is and is not. This is an *identification theorem*— $h_3(\mathbb{O})$ ’s cubic norm *is* the exceptional GST prepotential, so the matter–gravity couplings are determined by $\det(X)$ —conditional on the shared assumption that the algebraic spacetime carries smooth field-theoretic dynamics on a manifold. It is *not* an independent derivation of GR. The earlier lattice/Jacobson route was abandoned: the lattice is a modeling choice, and an attempted Weinberg-style forcing of $-R/2$ failed (the soft-graviton hypotheses are necessary, not sufficient). Smooth-manifold dynamics is the principal open bridge.

5.2 The Standard Model from $h_3(\mathbb{O})$

Paper 7 [Ehrlich, 2026g] argues that the Standard Model gauge group emerges from the exceptional Jordan algebra $h_3(\mathbb{O})$, the unique non-composable self-modeling structure. The argument is structural, not spectral-theoretic.

Why $h_3(\mathbb{O})$. Self-modeling forces Jordan algebras (Step 3 of the chain). The question is which Jordan algebra governs the universe as a whole. The answer follows from a purely algebraic argument whose key premise is radical relativity itself.

Claim 2 (Non-composability). *The universe’s self-modeling algebra is $h_3(\mathbb{O})$, conditional on the anthropic observer-grade restriction of Paper 7.*

Argument.

1. Self-modeling forces a Jordan algebra (Paper 5 [Ehrlich, 2026e], theorem).
2. Special (composable) Jordan algebras admit composites with nontrivial systems: for any special J , a well-behaved composite $J \otimes K$ exists [Barnum et al., 2020, Hanche-Olsen, 1983].
3. Under radical relativity (Postulate 2), all consistent mathematical structures exist. If composites of J exist mathematically, they exist. Therefore J is a subsystem of those composites.
4. Define the *universe* as the maximal self-modeling structure containing the observer. (Maximality is well-defined because assumption A1 restricts to finite dimension, bounding the Jordan-algebraic dimension from above.)
5. If the universe’s algebra U were composable (special), then U embeds in an associative C^* -algebra A [Hanche-Olsen, 1983]. Composites $A \otimes B$ exist for nontrivial B (by steps 2–3), and the composite $A \otimes B \cong M_{nk}(\mathbb{C})$ is again a C^* -algebra. By Paper 5’s main result [Ehrlich, 2026e], any $M_m(\mathbb{C})^{sa}$ admits a faithful self-model (take $V_M = V_B$, $\varphi = \text{id}$). Therefore the composite carries its own faithful self-modeling structure, strictly larger than U , in which the observer is embedded. This contradicts maximality of U .
6. Therefore the universe’s algebra is non-composable (contrapositive of step 5).
7. The unique non-composable simple Jordan algebra is $h_3(\mathbb{O})$ (JvNW classification [Jordan et al., 1934] plus Zel’manov [Zel’manov, 1979]).

Remark 5. Under radical relativity, algebraic composability *is* the physical question of subsystem-hood. There is no separate “physical” layer in which composability might differ from the algebraic property. This is why the argument uses no quantum entanglement or physical reasoning — only the Jordan-algebraic notion of composability, which is prior to quantum mechanics in the derivation chain. Step 7 requires simplicity (the universe’s algebra is not a direct sum); this is one of Paper 5’s four standing assumptions (simple EJA via trace-form non-degeneracy), an honest cost recorded in Sec. 6.4.

Remark 6 (Converse direction). Step 5 of the non-composability argument uses the converse of Paper 5’s main theorem: every $M_n(\mathbb{C})^{sa}$ admits a faithful self-model. The construction is explicit: take $V_M = V_B = M_n(\mathbb{C})^{sa}$, $\varphi = \text{id}$, and equip the composite with the standard quantum tensor product $M_n(\mathbb{C})^{sa} \otimes M_n(\mathbb{C})^{sa} = M_{n^2}(\mathbb{C})^{sa}$. For complex matrix algebras, the minimal and maximal composites coincide [Barnum et al., 2020], so the minimal-composite assumption (A3) is automatically satisfied. Together with Paper 5’s forward direction (self-modeling forces $M_n(\mathbb{C})^{sa}$), this gives: *a finite-dimensional spectral order-unit space admits a faithful self-model if and only if it is $M_n(\mathbb{C})^{sa}$ for some $n \geq 2$.*

The gauge group. $\text{Aut}(h_3(\mathbb{O})) = F_4$. An observer, modeled as a C^* -subsystem (Paper 5 [Ehrlich, 2026e]), picks a rank-1 idempotent $E_{11} \in h_3(\mathbb{O})$, inducing a Peirce decomposition into three sectors: V_1 (observer, 1-dim), $V_{1/2}$ (interface, 16-dim), and V_0 (complement, 10-dim). The stabilizer of E_{11} in F_4 is $\text{Spin}(9)$. A second choice — a complex structure $u \in S^6 \subset \text{Im}(\mathbb{O})$ — breaks F_4 differently, stabilizing $[\text{SU}(3) \times \text{SU}(3)]/\mathbb{Z}_3$. The intersection of these two stabilizers is the Standard Model gauge group $(U(1) \times \text{SU}(2) \times \text{SU}(3))/\mathbb{Z}_6$. This is the Todorov-Drenska theorem [Todorov and Drenska, 2018].

Chirality. The same complex structure u that determines the gauge group simultaneously determines the chiral representation. The Clifford algebra $\text{Cl}(6)$, generated by the six

imaginary octonions orthogonal to u , admits a Witt decomposition whose volume form produces an automatic left-handed embedding of Standard Model fermions. One choice does three things: gauge group, complexification, and left chirality. This unifies the Todorov-Drenska route (gauge group) with the Furey route [Furey, 2018] (chirality) via a single algebraic input.

Gap C. The argument that the observer’s C^* -algebra nature induces complexification of the Peirce half-space ($\text{Spin}(9) \rightarrow \text{Spin}(10)$) is physical reasoning, not a theorem. This is Gap C, the one remaining real gap in the SM derivation. Two independent motivations exist (algebraic and thermodynamic), but neither constitutes a proof. The honest assessment: Gap C may admit only an evolutionary resolution — “no chirality $\rightarrow$ no electroweak symmetry breaking $\rightarrow$ no mass hierarchy $\rightarrow$ no chemistry $\rightarrow$ no self-modelers” — the same status as “why eyes.” Overwhelmingly plausible; not a theorem; perhaps not closeable as one (Sec. 6.4).

Claim 3 (Uniqueness of ρ_J). *Within $h_3(\mathbb{O})$, the heuristic density (1) resolves uniquely. Paper 1 [Ehrlich, 2026a] proves: $\rho_J(X) = \det(X) \cdot (\text{Tr}(X^2) - \frac{1}{3})$ is the unique lowest-degree F_4 -invariant polynomial on the state space of $h_3(\mathbb{O})$ that vanishes at rank-deficient states ($\det = 0$) and at thermal equilibrium ($\text{Tr}(X^2) = \frac{1}{3}$), and is non-negative throughout.*

The cubic norm $\det(X) = \lambda_1 \lambda_2 \lambda_3$ is the exceptional invariant of $h_3(\mathbb{O})$ — it does not exist for ordinary quantum mechanics ($h_n(\mathbb{C})$). $\rho_J > 0$ requires the exceptional structure. The purity factor $\text{Tr}(X^2) - \frac{1}{3}$ measures distance from thermal equilibrium. Their product is the unique minimal F_4 -invariant *self-reference density* (the Observable-side weight in the phenomenological measure Φ of Paper 1, not experience itself): (three-way correlation) $\times$ (thermodynamic disequilibrium).

Remark 7 (The cubic prediction). ρ_J depends on $\det(X)$, a cubic invariant that vanishes when any Peirce sector is empty. Integrated information Φ [Tononi, 2008] is quadratic (pairwise integration), remaining nonzero when one sector is disrupted. The prediction: disrupting one of three sectors produces a sharp loss of self-modeling quality ($\det \rightarrow 0$), not a gradual reduction. This is the program’s most distinguishing falsifiable prediction.

5.3 Three generations from the Peirce tower [Speculative]

The Standard Model has three generations of fermions with a steep mass hierarchy ($m_t/m_u \sim 10^5$). Boyle [Boyle, 2020] noted that $h_3(\mathbb{O})$ contains three off-diagonal octonionic entries (x_1, x_2, x_3) permuted by $\text{SO}(8)$ triality, providing a structural reason for exactly three generations. Triality treats all three symmetrically and does not explain the mass hierarchy.

The iterated Peirce decomposition breaks this symmetry. Decomposing $h_3(\mathbb{O})$ under E_{11} gives $V_1 + V_{1/2} + V_0$, where $V_{1/2} = \mathbb{O}^2$ contains x_2 and x_3 but not x_1 . The complement $V_0 = h_2(\mathbb{O})$ is itself a Jordan algebra; decomposing it under a rank-1 idempotent E_{22} reveals x_1 as the interface at the second level. The tower bottoms out at the third level ($h_1(\mathbb{O}) = \mathbb{R}$). Interface dimensions descend: 16, 8, 1.

The observer’s existence breaks triality to a $2 + 1$ split: x_2 and x_3 are direct interfaces (level 1), while x_1 is one level removed inside the complement (level 2). Coupling strength to the observer — and therefore mass — decreases with Peirce depth. The third generation (heaviest) couples at level 1; the first generation (lightest) couples at level 2.

Remark 8. This is explicitly speculative. Whether the Peirce tower reproduces the quantitative mass hierarchy (the actual ratio m_t/m_u) depends on the spectral action on $h_3(\mathbb{O})$, which has not been computed. The structural prediction is qualitative: the hierarchy exists and has a $2 + 1$ pattern. See Sec. 7.3 for the status of the spectral action computation.

6 Objections

6.1 “Level IV is unfalsifiable”

Level IV itself — the claim that all consistent mathematical structures exist — is not directly falsifiable. Neither is the negation. The claim “only one structure is physical” is equally immune to direct test.

What *is* falsifiable is the derivation chain. If self-modeling forces quantum mechanics (Steps 3–6), then discovering a classical self-modeling system with equal predictive power would refute the framework. If the non-composability argument gives $h_3(\mathbb{O})$, then discovering a consistent non-special Jordan algebra besides $h_3(\mathbb{O})$ would break the uniqueness claim (though the classification theorem makes this impossible). The specific form of ρ within $h_3(\mathbb{O})$ (Claim 3) is now a theorem, with falsifiable predictions (Sec. 7.2).

The philosophical premise is not falsifiable. Its consequences are. This is the same epistemic status as any foundational principle in physics: the principle of least action is not falsifiable, but Lagrangian mechanics is.

6.2 “Genericity is measure-dependent”

The worry: “generic among what? With which measure?” If the derivation chain relied on typicality arguments — “most structures are quantum” — this would be fatal. It does not. The chain consists of *theorems*, not statistical claims.

Self-modeling forces Jordan algebras (van de Wetering). Local tomography forces complex type (dimension counting). Non-composability forces $h_3(\mathbb{O})$ (JvNW + Zel’manov). At no point does the argument appeal to “most structures have property X .” The word “generic” appeared in early versions of the program (specifically in the non-commutativity argument [Motzkin and Taussky, 1955]) but has been superseded. Paper 5’s chain is forcing, not typicality.

6.3 “This is anthropic reasoning”

Anthropic reasoning selects from a pre-existing landscape: “we observe quantum mechanics because only quantum-mechanical universes support observers.” This requires a landscape (where does it come from?) and a selection mechanism (what does the selecting?).

For Steps 3–6 (QM), the framework *derives*: self-modeling forces quantum mechanics by theorem. There is no landscape to select from. The algebraic structure is forced, not filtered. The distinction is between “only habitable houses are inhabited” (anthropic) and “this is the only house that can be built on this foundation” (forcing).

We acknowledge that Gap C (complexification, Sec. 5.2) *does* use anthropic reasoning: “no chirality $\rightarrow$ no chemistry $\rightarrow$ no self-modelers.” The program is not uniformly non-anthropropic. Steps 3–6 are forcing; Gap C is selection. This is registered honestly.

6.4 “The gaps are too big”

Honest assessment: the gaps are real. Table 2 registers every open gap in the program.

Table 2: Gap register. Gaps are classified by severity and shared status (whether other programs face the same gap).

Gap	Severity	Status
Smooth-manifold dynamics	Real	GST identification, not a derivation
Gap C (complexification)	Real	Evolutionary argument; not a thm
Minimal composite assumption	Significant	Load-bearing; not derived
Finite-dimensional assumption	Moderate	Natural for subsystems

The GR step is now an *identification* with the exceptional GST entry, not a lattice derivation; its open bridge is smooth-manifold dynamics, shared by every program that builds gravity on an algebraic or discrete substrate.

Gap C is the one genuinely novel gap. The complexification $\text{Spin}(9) \rightarrow \text{Spin}(10)$ is argued on physical grounds (the observer’s C^* -nature) but not proved. What would close it: a theorem of the form “ C^* -measurement maps on a real Jordan module necessarily factor through $\otimes_{\mathbb{R}} \mathbb{C}$.” In the absence of such a theorem, the evolutionary argument (“no chirality $\rightarrow$ no chemistry $\rightarrow$ no self-modelers”) provides overwhelming plausibility without mathematical certainty.

The minimal composite assumption does significant work in the local tomography proof (Step 4). All four adversarial reviewers of Paper 5 flagged this. It is an identified cost, not a hidden one.

6.5 “How is this different from Tegmark?”

Table 3 compares the present framework against related programs.

Tegmark: elevates mathematics to the status of the physical (“the physical world IS a mathematical structure”). We deflate the physical to the mathematical. The direction matters: Tegmark inherits a measure problem (which structure is “the” physical one?); we do not.

Mueller: provides a Solomonoff-type measure on observer experiences but does not derive the algebraic structure of physics. Our framework is compatible with Mueller’s measure

Table 3: Comparison with related programs. “Derives QM” = quantum mechanics follows from the framework’s premises. “Derives SM” = Standard Model gauge group follows. “Measure” = the framework provides a measure on observer-instances. “Honest gaps” = open problems are explicitly registered.

Program	QM	SM	Gaps	Key difference
Radical Rel.	Thm	Cond.	Yes	Forcing, not selection
Tegmark MUH	No	No	Part.	Elevates math to physical
Mueller Alg. Idealism	No	No	Yes	Measure only, no algebra
Wolfram Ruliad	Plaus.	Part.	No	“Can” not “must”
String Theory	Input	Landsc.	Part.	QM assumed, not derived
Barandes Stoch.-Q	Equiv.	No	Yes	Why unitary? (we answer)
Connes NCG	Input	Cond.	Yes	A_F chosen, not derived
Adler Trace Dyn.	Emergent	No	Yes	Assumes NC; we derive it

and would benefit from it as an input to structure space, but the derivation chain does not require it.

Wolfram: explores all possible computations (the “ruliad”) and claims that observers with certain properties will experience physics resembling ours. The claim is “can emerge,” not “must be.” Our chain is forcing: self-modeling *must* produce complex QM.

Barandes: establishes a bijection between unitary QM and indivisible stochastic processes. This is a load-bearing link in our chain (Step 7) but does not itself explain *why* unitary dynamics hold. Our chain provides the “why”: self-modeling forces C^* -algebras, whose automorphisms are unitary.

Connes: derives the Standard Model Lagrangian from a spectral triple, but the finite algebra $A_F = \mathbb{C} \oplus \mathbb{H} \oplus M_3(\mathbb{C})$ is chosen to match observation. Our framework derives $h_3(\mathbb{O})$ from non-composability and extracts the SM gauge group as the intersection of two stabilizers. The algebra is forced, not chosen.

7 The Research Program

7.1 What is explained, what is conditional, what is contingent

The program’s intellectual honesty requires distinguishing these categories sharply. “Forced” means the result follows by theorem from the premises. “Conditional” means the result holds subject to identified gaps that are registered explicitly. “Speculative” means a structural

Table 4: Status of physical content within the program.

Content	Status	Source
Complex QM	Forced*	Paper 5 (A1–A4)
Born rule	Forced	Gleason ($n \geq 3$)
Unitary dynamics	Forced	Skolem-Noether
$h_3(\mathbb{O}) =$ exceptional MESGT	Identification	Paper 6 (GST)
Einstein term $-R/2$	Identified	Paper 6 (smooth dyn. open)
SM gauge group	Conditional	Paper 7 (Gap C)
Chiral representation	Conditional	Paper 7 (Gap C)
3 generations	Speculative	Peirce tower
Mass hierarchy	Speculative	Peirce depth
Specific couplings	Contingent	Ring in structure space
3+1 dimensions	Plausible	Huygens + stability

argument exists but has not been formalized. “Contingent” means the result may not be uniquely determined by self-modeling.

Specific coupling constants, masses, and mixing angles may belong to the contingent category: the self-modeling condition constrains the algebra (A_F , forced) but not the dynamics (D_F , where parameters live). The viable parameter values likely form a manifold — a “ring around the basin” — not a unique point. The experiential measure ρ (Claim 3) selects a position on this ring, but the selection may have the same status as “why this brain?” rather than “why gravity?” 3+1 dimensions and the SM gauge group are likely universal across the ring (forced by algebraic structure). The fine structure constant may vary.

7.2 Falsifiable predictions

1. **No coupling unification (prediction).** The SM gauge group arises from an intersection of stabilizers in F_4 , not from a single GUT group breaking. The framework predicts that running couplings do *not* unify at a single scale.
2. **No Planck-scale modifications to QM (prediction).** Quantum mechanics is forced by the algebra of self-modeling (Steps 3–6), not by spacetime structure. There is no reason for QM to break down at the Planck scale. Searches for Planck-scale decoherence should find null results.

3. **Proton stability (consistency check).** No $SU(5)$ or $SO(10)$ GUT group appears in the framework. Proton decay via GUT-mediated channels is not predicted. This is consistent with current bounds ($\tau_p > 10^{34}$ years). Current data on coupling non-unification in the non-SUSY SM is also consistent with the intersection-of-stabilizers mechanism.
4. **Cubic vs. quadratic consciousness measure (prediction).** ρ_J (Claim 3) depends on $\det(X)$, a cubic invariant. IIT’s Φ [Tononi, 2008] is quadratic. Selectively disrupting one of three Peirce sectors should produce a sharp (not gradual) loss of self-modeling quality. Thalamocortical disruption data from anesthesia research can distinguish these functional forms.

7.3 Open problems

1. **Gap C (complexification).** The most pressing open problem. A theorem of the form “ C^* -measurement maps on a real Jordan module necessarily factor through $\otimes_{\mathbb{R}} \mathbb{C}$ ” would close it. Failing that, the evolutionary argument stands but the SM derivation remains conditional.
2. **Spectral action on $h_3(\mathbb{O})$.** Farnsworth [Farnsworth, 2015] has computed spectral actions on non-associative geometries, producing Einstein gravity coupled to G_2 gauge fields from octonionic spectral triples. Extending this to $h_3(\mathbb{O})$ would unify the GR route (Paper 6) with the SM route (Paper 7) under a single spectral action. The computation is open.
3. **Vanchurin dynamics.** Vanchurin [Vanchurin, 2020, 2024] showed that neural network learning dynamics reproduce aspects of quantum mechanics and general relativity. The present program contributes the learning objective: self-modeling. The loss function is $-I(B; M)$. Characterizing the fixed point of Vanchurin’s dynamics with this loss would address Level 6 questions (specific parameter values).
4. **3+1 dimensions.** 3+1 is the unique dimensionality where clean signal propagation (Huygens’ principle), stable atoms, stable orbits, and knot existence all hold simultaneously. The Huygens argument is specifically about self-modeling: a self-modeler receiving echoes of old signals double-counts information it already processed. Formalizing this as a channel capacity argument is open.
5. **Infinite-dimensional extension.** The full chain is proved for finite-dimensional spectral order-unit spaces. Quantum field theory is not finite-dimensional. The Bekenstein bound suggests that self-modelers in finite regions are necessarily finite-dimensional, but connecting this to the formalism is open work.
6. **Lean formalization.** Paper 5’s downstream derivation chain (EJA classification $\rightarrow$ local tomography $\rightarrow$ type exclusion $\rightarrow C^*$ -promotion) compiles in Lean 4 with Mathlib, with zero `sorry` placeholders and four axioms (three published results, one bridge axiom connecting the novel construction to the downstream machinery). Eliminating

the bridge axiom — proving S1–S7 for the corrected product directly in Lean — is in progress and would upgrade the entire chain to machine-verified.

7.4 Closing

The Copernican program has a simple logic: find a distinction that is doing no work, remove it, and see what follows. Each removal simplified the theory and expanded its explanatory scope. We have argued that the distinction between “mathematical” and “physical” is the last such distinction, and that removing it forces the physics observers experience — from the complex field to the Standard Model gauge group — through the algebraic consequences of self-modeling.

The chain is not complete. Gap C is open. The spectral action on $h_3(\mathbb{O})$ is uncomputed. The Vanchurin dynamics have not been solved with self-modeling as the loss function. But the chain is long enough, and the gaps are honest enough, to constitute a research program worth pursuing. The program’s distinguishing feature is not that it has fewer gaps than its competitors, but that it has *identified* gaps where others have hidden assumptions, and that its foundational premise — no privileged instantiation — is not an axiom added to physics but the last axiom removed from it.

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